LIGHT AND SOUND

Define these words

- Transparent

An object that allows most light to pass through it. Objects can be clearly seen.
eg. normal glass.

- Translucent

An object that allows the light to come through imperfectly
eg frosted glass

- Opaque

An object that does not allow any light to pass through
eg. a wall.

- Luminous

An object that releases its own light.
eg the sun

- Non-luminous

An object that reflects light.
eg the moon

- Reflection

Bouncing off the surface of a substance.

- Refraction

Change in speed of light as it passes from one substance to another.

- Dispersion

The seperation of white light into colours.

- Spectrum

A band of colours making up visible light.

- Pitch

highness or lowness of a sound. Depends on frequency.

- Frequency

number of vibrations in one second, or number of wavelengths passing in one second.

- Wavelength

Distance between 2 neighbouring crests or troughs of a wave.

- Decibels

a unit of measurement of relative sound intensity

Use the listed words to complete the sentences.

convex ✓	focal ✓	frequency ✓	microwaves ✓	opaque ✓
plane ✓	rays ✓	reflection ✓	scattered ✓	speed ✓
straight ✓	translucent ✓	transparent ✓	transverse ✓	

1. Light is a form of energy. Light travels in straight lines. The lines that are used to show the path of light are called rays.
2. Light travels at the speed of 300 000 kilometres per second in a vacuum.
3. Rays or beams of light may be seen when light is scattered or reflected from particles in substances and redirected to the eye.
4. Light reflected from a mirror follows the law of reflection which states: the angle of incidence equals the angle of reflection. Flat mirrors are called plane mirrors.
5. Curved mirrors may be concave (curved inwards) or convex (curved outwards). Light reflected from concave and convex mirrors also follows the law of reflection. Parallel rays of light are reflected to a focal point in a concave mirror. Convex mirrors reflect light from a wide angle creating an image reduced in size that captures much of the surrounds.
6. Shiny surfaces reflect light but if a surface absorbs light it is said to be opaque
7. If most of the light travels through a material, the material is said to be transparent Window glass is transparent.
8. Some materials allow some light to travel through but also scatter some of the light. Objects are not clearly visible when viewed through these materials. Frosty glass used in bathroom windows is an example. Such materials are said to be translucent
9. Light is an example of an electromagnetic wave. These waves are transverse waves. The distance from crest to crest is the wavelength of a wave.
10. The electromagnetic spectrum is divided into regions from low frequency to high frequency: radio waves; microwaves; infra-red waves; visible waves; ultra-violet waves; X-rays; gamma rays.
11. Analogue and digital television, wireless broadband and mobile phone calls can all be transmitted over long distances using high frequency radio waves, also called microwaves.

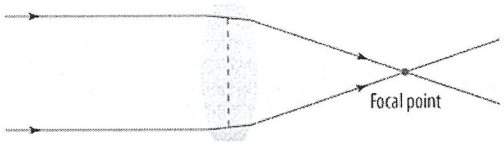
Complete the table below giving examples of materials that are;

Transparent	Translucent	Opaque
windows	frosted glass	books
glad wrap		desk
clear cellophane		walls.

12. Use the terms below to identify the lenses and mirrors shown.

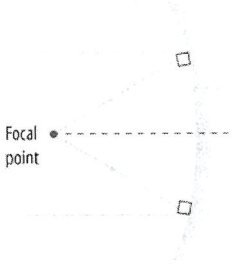
Concave mirror	Convex mirror	Converging lens	Diverging lens

(i)



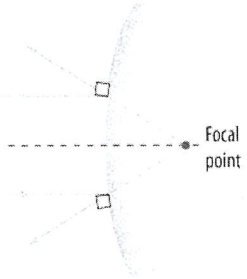
Convex/converging lens

(ii)



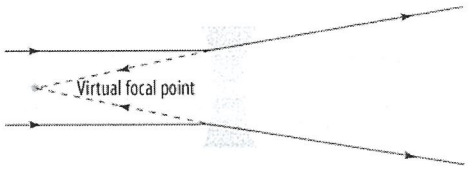
concave mirror

(iii)



convex mirror

(iv)



Concave/Diverging lens

b. Which of the mirrors or lenses above would you use as a magnifying glass?

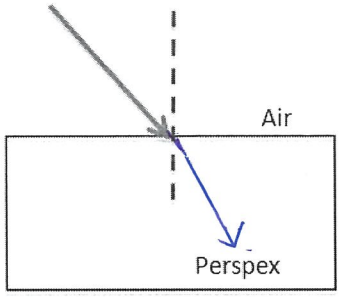
convex lens

c. Which of the mirrors or lenses above would you find being used at an intersection so that drivers can see around a blind corner?

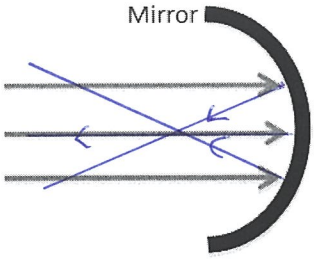
convex

13. For each of the diagrams below, draw the path of the light beams that would result.

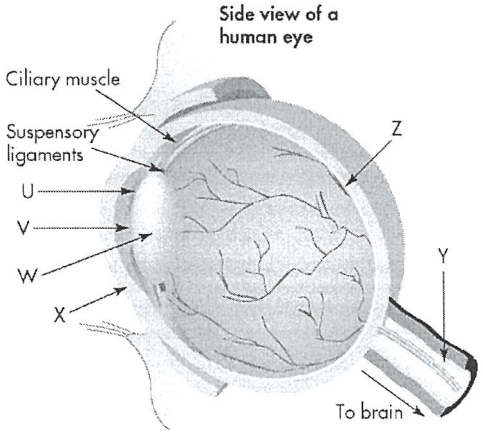
(a)



(b)



(a) Identify each of the structures labeled U–Z in the diagram of the eye below.



U: Iris

V: Pupil

W: lens

X: Cornea.

Y: Optic nerve

Z: retina

(b) Identify the structure of the eye that matches each of the following descriptions:

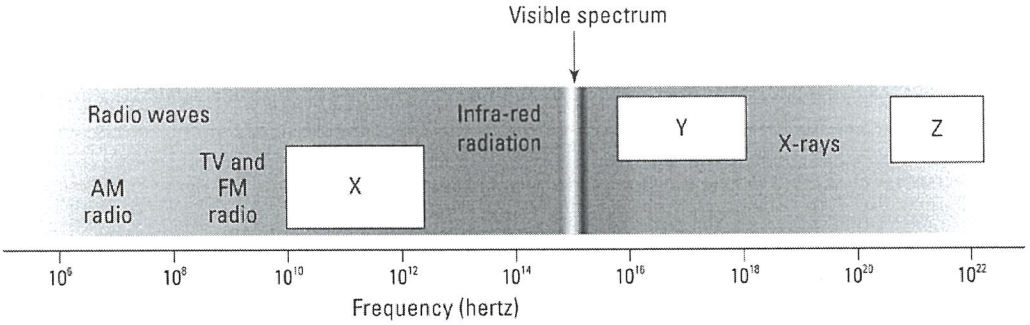
A. Bends the incoming light, so that it passes through the pupil towards the lens.

B. The image forms here. The light-sensitive cells detect the brightness and colour of the different parts of the image.

C. The opening through which light travels towards the lens.

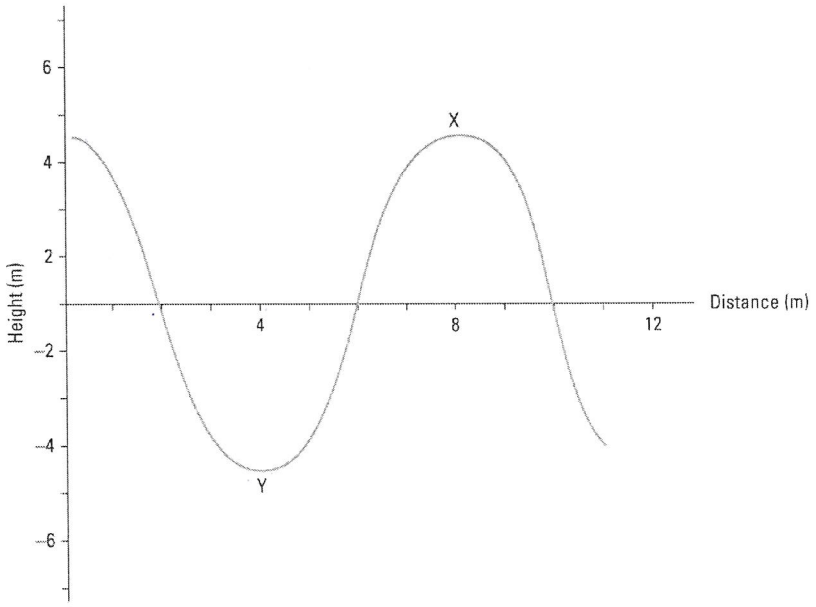
A: Cornea B: retina C: pupil

14. The following diagram shows the regions of the electromagnetic spectrum. Identify the missing regions X, Y and Z.



X: Microwaves + Radar
Y: Ultraviolet radiation
Z: Gamma rays.

15. Examine the following diagram of a wave.



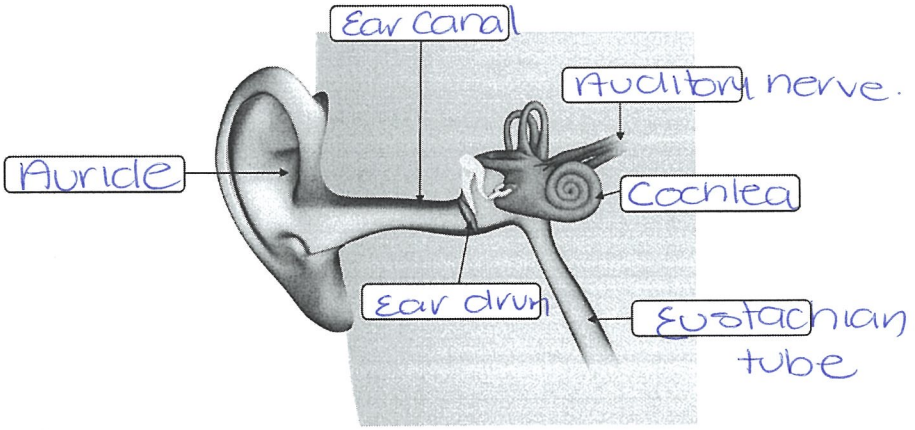
(a) Identify the type of wave illustrated. Transverse

(b) Two parts of the wave are labelled X and Y. Name these parts.
X: crest Y: trough

(c) Use the scale to determine the wavelength and amplitude of the wave.
Wavelength: ~8m
Amplitude: ~4.5m

16. Insert the following words into the diagram of the auditory system below.

Cochlea ✓	Eustachian tube ✓	Ear drum ✓
Auricle ✓	Auditory nerve ✓	Ear canal ✓



Looking Back.
2nd ed pg 350 q 1, 2, 5, 6, 13, 14, 16, 17,